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Social Influence Analysis in Social Networking Big Data: Opportunities and Challenges

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Abstract

Social influence analysis has become one of the most important technologies in modern information and service industries. It will definitely become an essential mechanism to perform complex analysis in social networking big data. It is attracting an increasing amount of research ranging from popular topics extraction to social influence analysis, including analysis and processing of big data, social influence evaluation, influential users identification, and information diffusion modeling. We provide a comprehensive investigation of social influence analysis, and discuss the characteristics of social influence and the architecture of social influence analysis based on social networking big data. The relationship between big data and social influence analysis is also discussed. In addition, research challenges relevant to real-world issues based on social networking big data in social influence analysis are discussed, focusing on research issues such as scalability, data collection, dynamic evolution, causal relationships, network heterogeneity, evaluation metrics, and effective mechanisms. Our goal is to provide a broad research guideline of existing and ongoing efforts via social influence analysis in large-scale social networks, and to help researchers better understand the existing work, and design new algorithms and methods for social influence analysis.

Introduction

Social networking is used widely as an important communication media with exponential growth. Specifically, Web 2.0 technology has brought revolutionary changes to our daily lives in online social networks. In the last decade, lots of online social networks, such as Facebook, Twitter, and LinkedIn, have emerged and tightly connected web users all over the world. People can directly engage in these networks, then build their own friendship networks, and share their opinions, insights, information, experiences, and perspectives with each other. In addition, they can discover and spread information by using various formats such as words, pictures, audio, and video.

Social networking big data is a collection of very huge data sets with a great diversity of types from social networks (e.g., Facebook, Twitter, and LinkedIn). The 5V characteristics of social networking big data are volume, velocity, variety, value, and veracity, which make it difficult to handle such big data using traditional tools and techniques. Nowadays, the explosion of data in terms of high volume, high velocity, and high variety, fueled by stunning and exciting advances in information technologies and web techniques, has become the focus of widespread attention. Big data applications [1] lie in many scientific disciplines, such as physics, astronomy, atmospheric science, medicine, genomics, biology, biogeochemistry, and other complex and interdisciplinary areas of scientific research.

Social influence analysis [2] is pervasive throughout society. The high-level goal of social influence analysis is to answer questions related to social influence, such as "Who can be influenced?," "Who can influence whom?," "Who are vulnerable to influence?" "Why is a user attracted to a particular group?," and "Who are the most influential users in a specific social network?" The main idea of social influence analysis is how to quantify the influence of each user and how to identify the most influential users in social networks. It has great potential to help us understand the ways in which information, experiences, ideas, and innovations propagate across social networks. Analyzing and mining social networks can provide new insights into how people interact with and influence each other, and why their ideas and opinions on different subjects can spread in social networks.

Analysis and modeling on social influence is becoming an important part of research on social networks. By analyzing the influencing mode among users and the spreading mode of influence, based on social networking big data, the following advantages can be obtained: understanding of social behaviors of people from the angle of sociology, a theoretical basis for making public decisions and public opinion guidance, and promoting communication and dissemination of political, economic, and cultural activities, as well as in other fields. Thus, social influence analysis in social networks has important social significance and application value using social networking big data.

The goal of this article is to provide a comprehensive investigation of the status of social networking big data for social influence analysis, and the characteristics and architecture of social influence analysis in social networking big data. The relationship between social influence analysis and big data is discussed, and the opportunities for social influence analysis are also discussed. In addition, research challenges are provided, focusing on research issues such as scalability, data collection, dynamic evolution, causal relationship, network heterogeneity, evaluation metrics, and effective mechanisms.

The remainder of this article is organized as

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